ETC5860 Research Proposal

Assessing the Robustness of CD-type Tests During Stress Episodes

Adrian Lim

Supervised by: Natalia Bailey & Bonsoo Koo

Research Question and Motivation

Research Questions

Q1 Do standard CD-type (cross-dependence-type) tests remain valid under non-linear dependence?

Q2 How does common volatility effects affect the finite-sample performance of CD-type tests?

Q3 Can robust self-normalized statistics improve inference in panel data CD testing?

Motivation

Dependence may be non-linear

- Cross-sectional units may share **common volatility dynamics** despite having zero correlation

Cross-sectional dependence affects inference

- Misspecifying the dependence structure invalidates standard inference procedures

Existing tests rely on strong assumptions

- Standard tests for cross-sectional dependence in panel data are **derived under i.i.d. assumptions**

Need for robust panel diagnostics

- Empirical applications for panel data require test procedures robust to heterogeneity and conditional heteroskedasticity

CD-Type Tests for Cross-Sectional Dependence

Developments in CD-Type Tests - Comparison Table

Test	Based on	Improvement	Limitation
LM, Scaled LM	Squared pairwise cross-correlations	Classical benchmark	Invalid when T is fixed N is large
CD, CD(p)	Pairwise cross-correlations	Correct size for fixed N, T	Biased for models with multifactor error structure
CD*, CD_W and CD_{W+}	Pairwise cross-correlations	Corrects multifactor bias	Does not address non-linear dependence

Key Limitation

Current CD-type tests are primarily designed for **linear dependence** and may become **invalid under conditional heteroskedasticity or broader forms of heterogeneity.**

Literature Gap: Robust Tests for Panel Data

Robust Correlation-Based Tests: Timeline

Taylor (1984); Cumby & Huizinga (1992)

- Robust t-type and Q-type (portmanteau) tests for zero serial correlation under heteroskedasticity.

Kyriazidou (1998)

- Robustified zero cross-correlation tests; assumed stationarity, no unconditional heteroskedasticity.

Dalla et al. (2022); Giraitis et al. (2024)

- Relaxed stationarity and unconditional heteroskedasticity; robust tests under heteroskedastic and dependent data.
- Broader class of uncorrelated data with general heterogeneity — including non-smooth and stochastic heteroskedasticity.

Limitation of Existing Robust Procedures

- Focus on univariate/bivariate settings
- → Not designed for panel CD diagnostics
- → Leaves robustness under nonlinear panel dependence unresolved

Research Objective

Aim

Goal: To adapt the results of Dalla et al. (2022) and Giraitis et al. (2024) to CD-type statistics for testing cross-sectional dependence in panel data models to produce tests that are robust under conditional heteroskedasticity and general heterogeneity, thereby enhancing their applicability in empirical work.

Contributions

1. Simulation studies assessing the finite-sample performance of CD-type statistics under non-linear dependencies
2. Robust corrections following Dalla et al. (2022) and Giraitis et al. (2024) in the event of test over-rejection in uncorrelated panels
3. Application of robust tests to real house price changes across Metropolitan Statistical Areas in the United States for evidence of non-linear dependence

The Problem Formally

Under I.I.D. Errors:

$$u_{it} = \sigma_i \varepsilon_{it}, \qquad \varepsilon_{it} \sim i.i.d.(0,1)$$

Then: $\mathrm{E}(CD) = 0, \qquad CD \to^{d} N(0,1)$

Under Non-Linear Dependence:

$$u_{it} = \sigma_{it} \varepsilon_{it}, \qquad \varepsilon_{it} \sim i.i.d.(0,1)$$
$$\sigma_{it} = \lambda_i h_t, \qquad h_t^2 \sim \text{GARCH}(1,1)$$

Then: $\mathrm{E}(u_{it} u_{jt}) = 0, \qquad Cov(u_{it}^2, u_{jt}^2) = f(h_t^2) \neq 0$

i.e., the standard CD statistic will still be correctly centered, but its limiting distribution depends on h_t^2 and may not be standard normal → size distortions

Standard CD Statistic

$$CD = \sqrt{\frac{2T}{N(N-1)}} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \hat{\rho}_{ij}$$

$$\hat{\rho}_{ij} = \frac{\sum_{t=1}^{T} e_{it} e_{jt}}{\left(\sum_{t=1}^{T} e_{it}^2\right)^{1/2} \left(\sum_{t=1}^{T} e_{jt}^2\right)^{1/2}}$$

Robust Self-Normalized Statistic by Dalla & Giraitis et al.

$$\tilde{t}_{ij,k} = \frac{\sum_{t=k+1}^{T} e_{it} e_{j,t-k}}{\left(\sum_{t=k+1}^{T} e_{it}^2 e_{j,t-k}^2\right)^{1/2}}$$

i.e., the specific second-moment normalization ensures that the statistic will be asymptotically normal even under general forms of heterogeneity.

Monte Carlo Design

Data-Generating Process

$$u_{it} = \gamma_i f_t + \sigma_{it}\varepsilon_{it}$$

Specify a multifactor error structure f_t to capture linear dependence and a common GARCH-type dynamic in σ_{it} to introduce nonlinear dependence

Null Hypothesis & Rejection Rule

$$H_0\colon \gamma_i = 0 \ \forall i, \qquad CD_{stat} > c_\alpha$$

Assess rejection rates under null hypothesis of no linear dependence

Simulation Objective: Isolate the effect of nonlinear dependence on the finite-sample validity of CD-type statistics and adjust the statistic accordingly.

Preliminary/Baseline Results

Table 2: Size and Power of CD test under $\chi^2(1)$ errors

T/N	5	10	20	30	50	100
Size $\gamma_i = 0$						
5	0.068	0.064	0.077	0.059	0.070	0.065
10	0.056	0.057	0.056	0.062	0.047	0.059
20	0.052	0.055	0.051	0.047	0.047	0.033
30	0.052	0.054	0.055	0.042	0.049	0.043
50	0.048	0.048	0.043	0.040	0.052	0.056
100	0.049	0.042	0.049	0.050	0.046	0.051
Power $\gamma_i \sim iid\, U(0.1, 0.3)$						
5	0.142	0.199	0.419	0.616	0.731	0.878
10	0.115	0.227	0.516	0.746	0.909	0.981
20	0.144	0.393	0.562	0.835	0.980	1.000
30	0.108	0.349	0.813	0.917	0.994	1.000
50	0.305	0.519	0.807	0.991	1.000	1.000
100	0.525	0.707	0.988	1.000	1.000	1.000

Table 3: Size and Power of CD test under time-varying unconditional heteroskedasticity

T/N	5	10	20	30	50	100
Size $\gamma_i = 0$						
5	0.093	0.064	0.066	0.061	0.065	0.065
10	0.060	0.053	0.054	0.056	0.047	0.058
20	0.031	0.044	0.050	0.050	0.046	0.051
30	0.045	0.042	0.047	0.047	0.053	0.047
50	0.046	0.051	0.048	0.052	0.052	0.051
100	0.042	0.045	0.061	0.047	0.053	0.060
Power $\gamma_i \sim iid\, U(0.1, 0.3)$						
5	0.094	0.115	0.155	0.356	0.458	0.694
10	0.079	0.139	0.320	0.400	0.491	0.859
20	0.083	0.191	0.351	0.521	0.755	0.963
30	0.086	0.169	0.367	0.654	0.802	0.988
50	0.058	0.135	0.395	0.662	0.947	1.000
100	0.090	0.210	0.704	0.924	0.999	1.000

Purpose: Replicate the finite-sample results of the standard CD test following Pesaran (2004, 2021) before introducing nonlinear dependence to assess potential over-rejection.

Results: Correct size following nominal 5% significant level under chi-squared errors and time-varying unconditional variance.

Empirical Application of the Robust Tests

Dataset Summary

Data

Real house price changes across Metropolitan Statistical Areas in the United States

Period

2000Q1 – 2012Q4

Observations

52 quarterly observations

Relevance

- Major disruptions or “stress periods” generate common volatility shocks
- These effects may remain even after factor filtering
- Ideal setting for robust CD diagnostics

Study Period Note: The 2000Q1–2012Q4 period includes the U.S. housing boom and the Global Financial Crisis, providing a natural setting for examining common volatility dynamics and nonlinear cross-sectional dependence

Under Strong Factor Dependence: Additional bias-correction procedures are considered when strong latent factors are present..

Timeline of Project

Month	Task
June	Perform Simulations
July	Apply to real-world data
August	Analyse results
September	Write thesis
October	Finalise thesis